

TrES-2b

R-2351

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110403 · created 2026-08-19T01:18:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455654.947003 +/- 7.3e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.013
Transit depth (Rp/Rs)^2	2.07 +/- 0.37 [%]
Orbital Inclination (inc)	83.61 +/- 0.39
Airmass coefficient 1 (a1)	1.0323 +/- 0.008
Airmass coefficient 2 (a2)	-0.0211 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	#2 - [384, 41]
Optimal Aperture	4.92
Optimal Annulus	8.6
Transit Duration (day)	0.0723 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.013 vs 0.1254 (deviation 1.26σ)
Duration fit vs published	0.0723 d vs 0.0746 d (deviation 0.31σ)
Mid-time O–C	0.2 min
Depth SNR	9.8
Residual scatter	0.77 %

Light curve

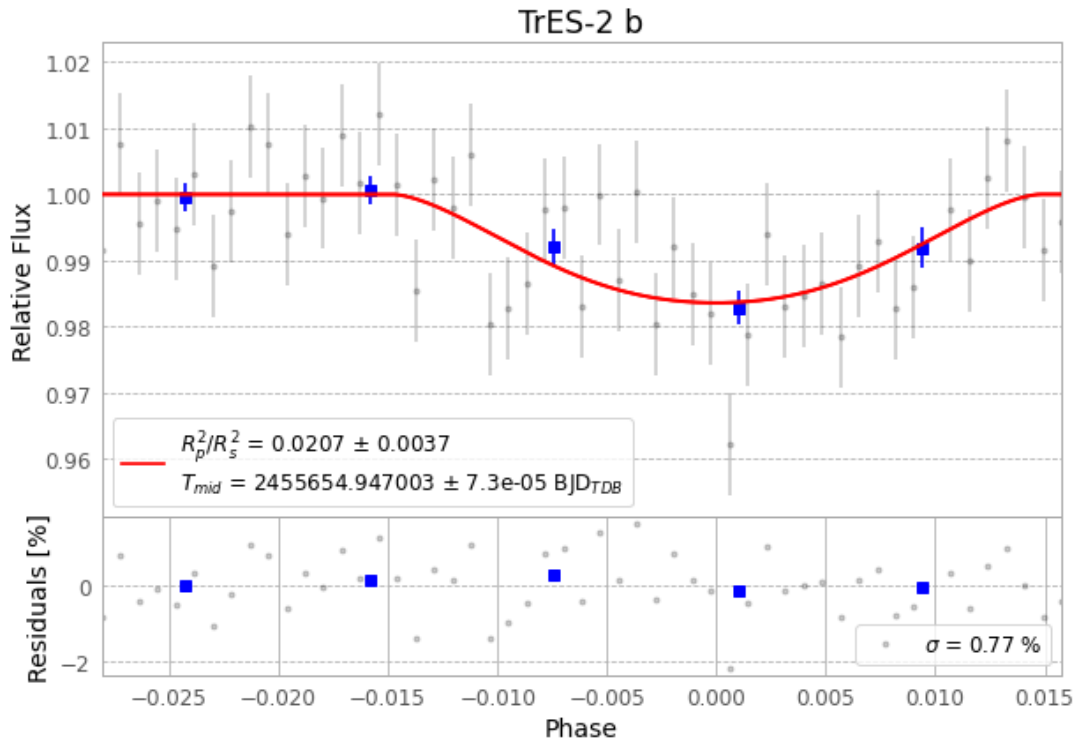


Figure 1 — FinalLightCurve_TrES-2 b_2011-04-03.png (SHA-256 345b00a07cd53e16...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 201], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 dff16bffe73129e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

53 FITS frames (35 MB), TRES-2110403090312.FITS → TRES-2110403113913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110403.log (40d874588314...), FinalLightCurve_TrES-2 b_2011-04-03.png (345b00a07cd5...), FinalLightCurve_TrES-2 b_2011-04-03.csv (7cca55c07ea0...)

Compute resources

Wall time 159.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:18Z.